

화학공학개론 (BE204-01)

학기	2026년 2학기
담당교수	김운혁
이수구분	교선�필수
학점	3.0
강의시간	월10:30-12:00(E7 - 236), 수10:30-12:00(E7 - 236)
강의실	E6 312 (교수실)
소속	기초학부

교과개요

Chemical engineering is a multidisciplinary field that focuses on the economic production of chemical products through the application of chemical reactions. Beyond chemistry, it requires a comprehensive understanding of several key areas, including: Mass/Energy Balance, Reaction Engineering, Fluid Dynamics, Heat and Mass Transfer, Process Design and Control. This introductory course aims to make these concepts accessible and easy to understand. It provides a clear understanding of what chemical engineering entails and is designed to equip students with the fundamental skills needed to integrate and apply these principles effectively in real-world scenarios.

교과목표

To distinguish between chemistry and chemical engineering, and understand the role of a chemical engineer. To get essential chemical engineering concepts, including unit conversion, mass balance, energy balance, reaction engineering, fluid mechanics, heat and mass transfer, process design, and control. To learn how to interpret a given chemical process from an engineering perspective.

주차별 강의계획

1주차:

2주차:

3주차:Chapter 1: Mole Balances

4주차:

5주차:Chapter 2: Conversion and Reactor Sizing

6주차:

7주차:

8주차:Chapter 3: Rate Laws

9주차:

10주차:Chapter 4: Stoichiometry

11주차:

12주차:Chapter 5: Isothermal Reactor Design: Conversion

13주차:

14주차:Chapter 6: Isothermal Reactor Design: Moles and Molar Flow Rates

15주차:

16주차:

학습윤리

A sincere and proactive attitude is required.

교과정책(방침)

Some engineering calculators may be required.